SHUBHAM SAHU

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PROFESSIONAL SUMMARY

Electronics & Computer Engineering student (CGPA 8.10/10 through Sem III) building embedded firmware and robotics hardware: a ROS 2 waste-sorting robot with YOLOv8 vision, an ESP32 smart-insole gait platform, an ESP32-CAM security unit and a roll-and-fly rescue drone design. Seeking a hardware-focused internship from Dec 2026 or Summer 2027.

SKILLS

- **Embedded:** ESP32, Arduino (C/C++), sensor fusion (FSR, MPU6050), I2C, UART, PWM, servo control (PCA9685), WiFiManager captive-portal provisioning, esptool-js, STK500v1 bootloader protocol, Web Serial API
- **Robotics & Vision:** ROS 2 (rclpy, topics, node design), YOLOv8, OpenCV, PyQt5/QtWebEngine HMIs, PD control tuning
- **Software:** Python, C/C++, TypeScript, FastAPI, Flask, React, Supabase/PostgreSQL, WebSocket, IndexedDB
- **Tools & Methods:** Docker, GitHub Actions CI/CD, Git, Agile/Scrum, circuit design & troubleshooting

EXPERIENCE

Autonomous Robotics Intern — MakerWorks Lab, Mumbai *Jun 2026 – Jul 2026*

- Built R3Bin, an AI waste-sorting robot: full ROS 2 node graph — camera detection node → touchscreen HMI → servo-control node → ESP32 hardware bridge — communicating over ROS topics
- Integrated a YOLOv8 vision model classifying waste (plastic, paper, metal, landfill) in real time, with a human-in-the-loop correction flow logging every decision locally and to Supabase
- Built a hybrid PyQt5 + QtWebEngine touchscreen HMI bridging ROS 2 logic to an HTML/Tailwind UI via QWebChannel, and a hardware-free simulation harness (ros2 topic pub) to validate the full pipeline

Embedded Coding Intern — Fostride, Mumbai *Dec 2025 – Feb 2026*

- Developed and tested embedded firmware for IoT and automation hardware on ESP32; reusable driver modules reduced integration time ~20%
- Implemented sensor integration and serial protocols (I2C, UART), enabling real-time data pipelines across 3+ hardware setups

Head of Product Development — Robotics & Automation Club, TSEC Mumbai *Feb 2025 – Jul 2026*

- Led a 15-member cross-functional team through 3+ product builds from ideation to prototype, delivering every showcase on time

PROJECTS

NIVA (GaitGuard Nexus) — ESP32 Smart-Insole Gait Analytics (ESP32, FSR, MPU6050, React, Web Serial)

- Built a real-time, explainable biomechanics engine (gait phase, cadence, center-of-pressure, stability, anomalies) where every score traces to an auditable sensor-threshold rule — no black-box model
- Engineered dual telemetry ingestion (USB via Web Serial API, WiFi via WebSocket with HTTPS-safe relay fallback) and shipped ESP32 firmware with WiFiManager captive-portal setup for non-technical users
- Designed sensor-fusion hardware (FSR + MPU6050, 40 kHz I2C with 5 kΩ pull-ups) scoped to an under-₹3,000 BOM for Indian community health centers

SURANG-SUTRA — Roll-and-Fly Mine-Rescue Drone + Relay Pucks (Vishwakarma Awards 2026-27 Stage 2 shortlist; design)

- Designed a collision-tolerant scout: an X-quad inside a single-axis rotating geodesic cage that rolls on sound floor and flies over debris or water, plus a cassette of five communication-only relay pucks dropped before each bend to keep the operator link alive in dark, GPS-denied galleries
- Avionics and relay-puck KiCad schematics (ERC clean) with pin map, power budget and costed BOM; design stage only, every figure is a stated target or calculation

Aurigen — Browser-Based ESP32/Arduino IDE: Blockly → C++ → cloud arduino-cli compile → USB flash over Web Serial

- Implemented extension-free USB programming for two board families: ESP32 via esptool-js and Arduino Uno via a hand-rolled STK500v1 bootloader protocol over Web Serial
- Fixed a real bootloop bug by exporting all four ESP32 flash partitions (bootloader, partition table, boot_app0, app) for deterministic first boot on factory-fresh boards

SENTRI — ESP32-CAM Visitor Security System (ESP32-CAM, Flask, OpenCV, Next.js)

- Built the physical edge unit: ESP32-CAM streaming live video into a Flask + OpenCV console with real-time face-detection overlays and a servo that pans to track the visitor's face
- Designed an explainable 0–100 visitor trust-score engine with an Isolation Forest anomaly stack; validated ~₹600 per-unit hardware cost targeting India's 50,000+ gated communities

EDUCATION

B.E. Electronics & Computer Engineering — Thakur Shyamnarayan Engg. College (TSEC), Mumbai | CGPA 8.10/10 (Sem I-III)
Expected 2028

HONORS & CERTIFICATIONS

Vishwakarma Awards 2026-27 Stage 2 shortlist • Niva at Automation Expo (Jul 2026) and India Innovates 2026 • 2nd place, Monad Blitz Mumbai V4 • Winner, Lovable × Anthropic Buildathon • Top 10, M-Indicator AI Hackathon

Microsoft Certified: Azure AI Fundamentals (AI-900), Jan 2026 • Lean Six Sigma Yellow Belt — CSSC, Mar 2024